

ENGINEERING AI ACROSS BORDERS

Building AI for Real-World Global Challenges



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WE HAVE MORE AI THAN EVER BEFORE

Millions of AI Projects.

Thousands of AI Startups.

Billions Invested.

AI adoption is accelerating faster than ever —
yet most projects never create real impact.

A SIMPLE QUESTION

How Many AI Projects Actually Reach Production?

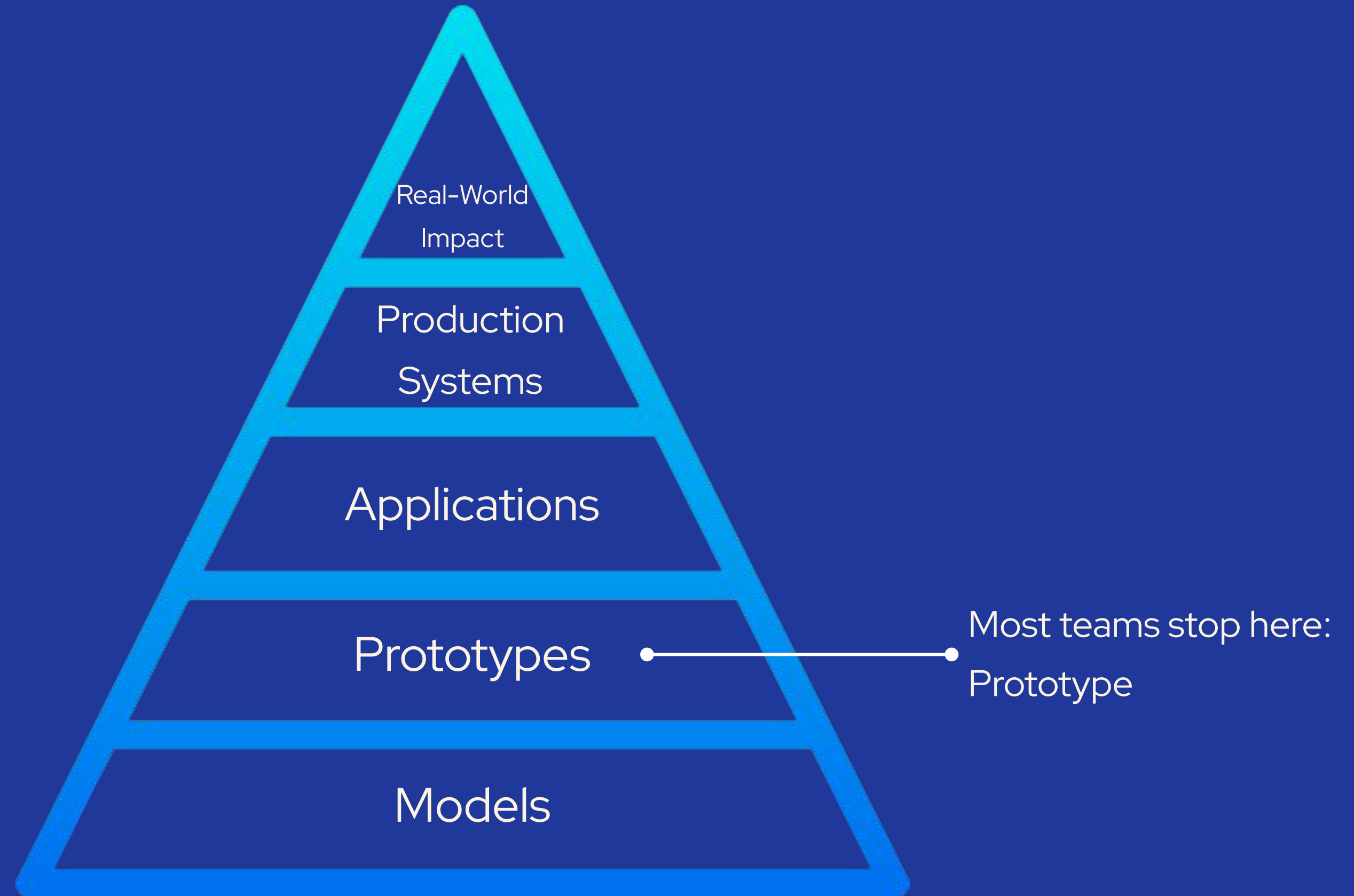
Raise your hand if you have built an AI prototype.

THE AI FUNNEL



"Building a demo is easy. Creating impact is hard."

THE AI IMPACT PYRAMID



THE BIGGEST MISCONCEPTION

AI Success $\neq$ Better Models

AI Success = Problem $\times$ Data $\times$ Workflow $\times$ Adoption $\times$ Governance

WHY PROJECTS FAIL



No Real
Problem



Poor Data
Quality



No Workflow
Integration



Lack of Trust



No Ownership

THE MISSING LAYER

Technology



People



Impact

IF AI IS NOT JUST ABOUT MODELS...

Then What Does It Take To Build AI Across Borders?

BORDERS / MISCONCEPTION

Most People Think Cross-Border AI Means Translation.

English

Chinese

Thai

Arabic

Spanish

Cross-Border AI is not Translation

BORDERS / DIMENSIONS

There Are Many Borders.

01

Language

02

Culture

03

Regulation

04

Infrastructure

05

Organization

BORDERS / 01 LANGUAGE

Border #1 — Language.

English $\neq$ Chinese $\neq$ Thai $\neq$ Arabic

Language is more than translation.

BORDERS / 02 CULTURE

Border #2 — Culture.

Trust

Communication Style

User Expectations

Decision-Making

The same AI may succeed in one country and fail in another.

BORDERS / 03 REGULATION

Border #3 — Regulation.

GDPR

PDPA

EU AI Act

Data Residency

AI must comply with very different legal environments.

BORDERS / 04 INFRASTRUCTURE

Border #4 — Infrastructure.

Cloud Availability

Connectivity

Device Capability

Compute Cost

Not every region operates in the same technology environment.

BORDERS / 05 ORGANIZATION

Border #5 — Organization.

Different Priorities

Different KPIs

Different Incentives

Different Owners

Many AI projects fail because organizations are not aligned.

BORDERS / TRANSITION

Why Cross-Border AI Is Hard.

Same AI
Different Countries
Different Results

United States

China

Thailand

Indonesia

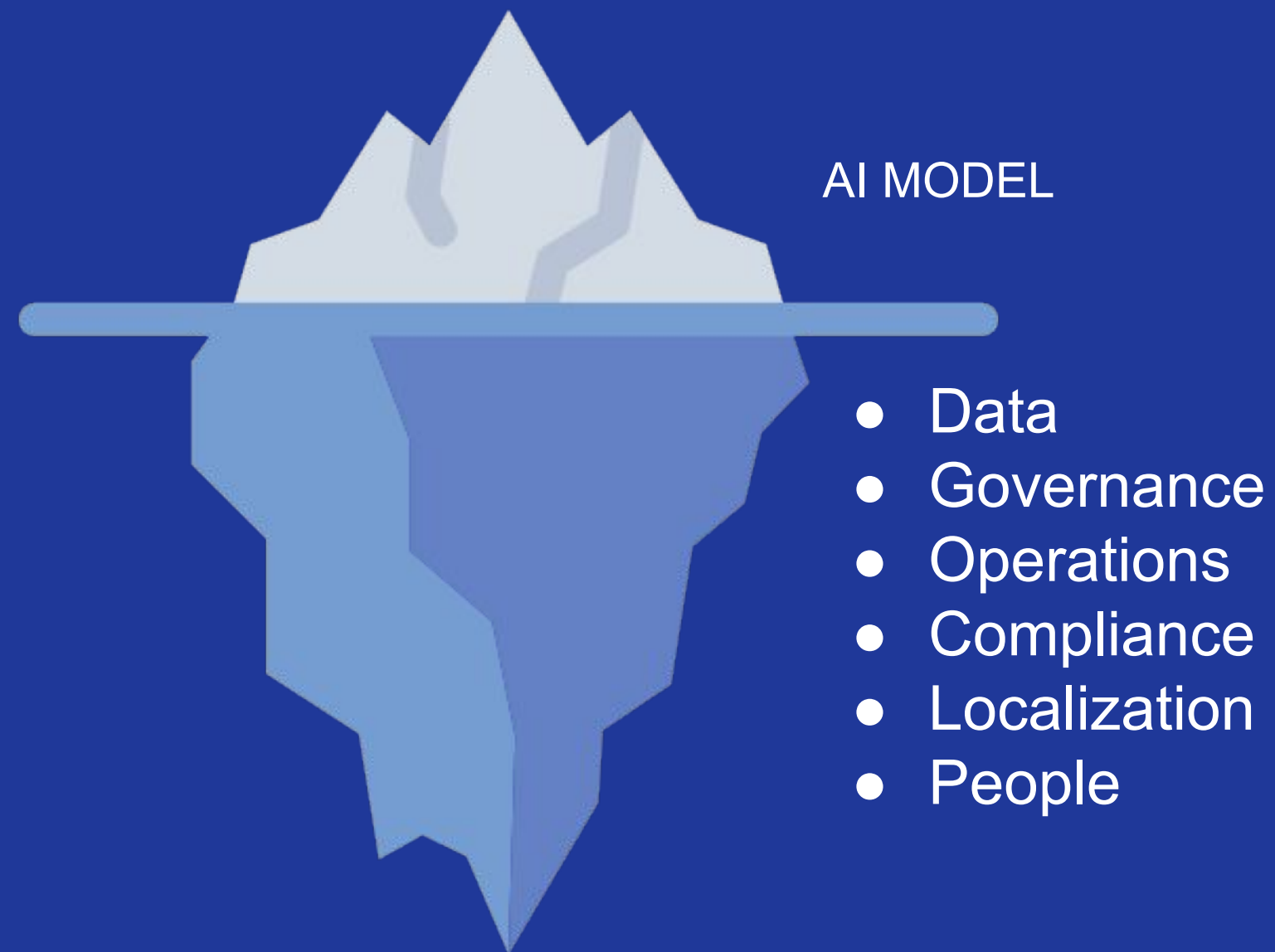
A Real Example.



BORDERS / COMPLEXITY

The Hidden Complexity.

Most complexity
is invisible.



BORDERS / FRAMEWORK

Engineering AI Across Borders.



Successful AI requires engineering across all five dimensions.

BORDERS / CHECKLIST

Before Building AI, Ask These Questions.

- Can users trust it?
- Can it scale?
- Can it comply?
- Can people adopt it?
- Can it survive reality?

NEXT - Designing AI Systems for Real-World Impact

ANATOMY / THE MISTAKE

The Biggest Mistake in AI Architecture



AI System = LLM

A model alone is not a system.



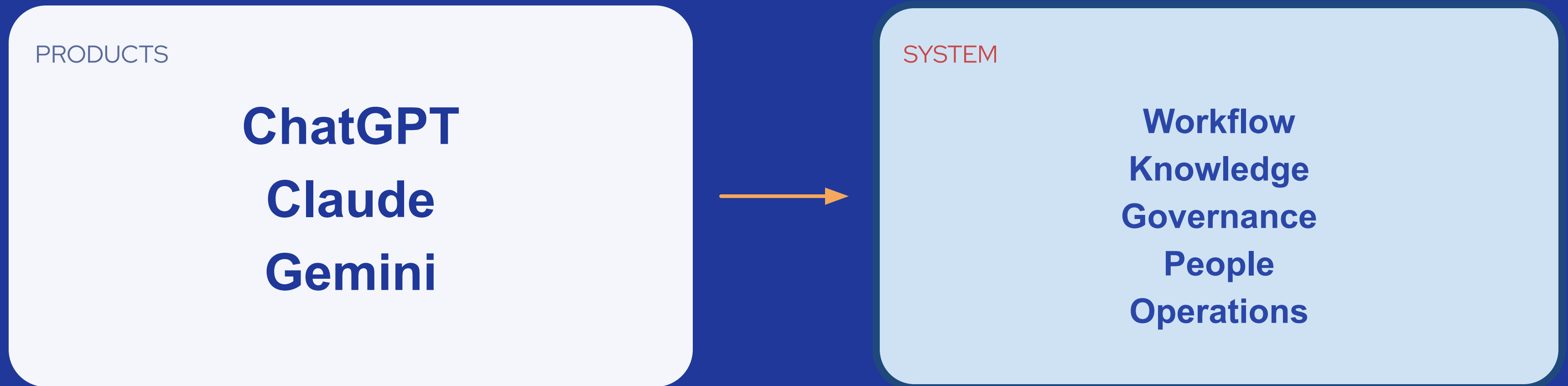
AI System =

People + Data + Workflow + AI + Governance

An AI system is much more than a model.

ANATOMY / PRODUCT VS SYSTEM

AI Is Not a Product. AI Is a System.



Choosing a model is easy. Designing a system is hard.

ANATOMY / ARCHITECTURE

What Does a Real AI System Look Like?

Layer 1 - Users

Mobile, web, chat, voice, API

Layer 2 - Applications

Workflows and business processes

Layer 3 - AI Services

LLMs, RAG, agents, tools

Layer 4 - Knowledge Layer

Documents, databases, APIs

Layer 5 - Enterprise Systems

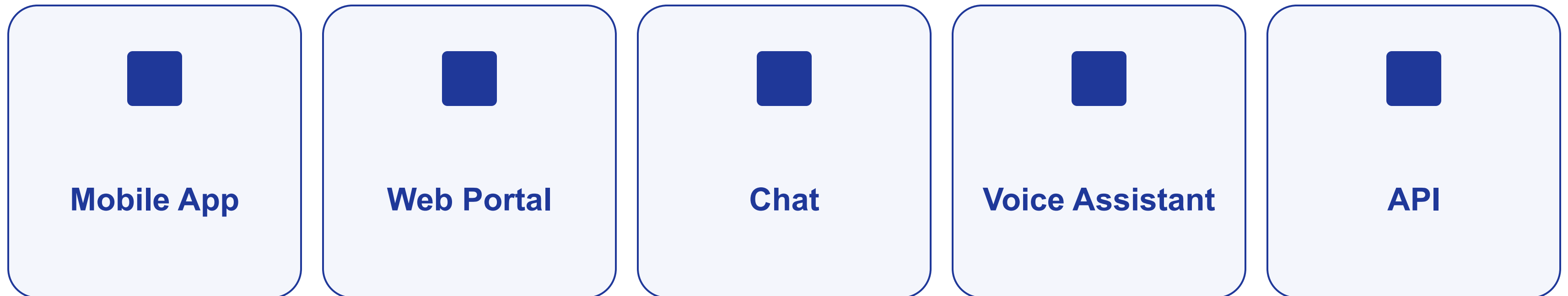
CRM, ERP, operational data

Layer 6 - Governance Layer

Security, compliance, monitoring

ANATOMY / LAYER 1

Layer 1 - Users



Users care about outcomes, not models.

ANATOMY / LAYER 2

Layer 2 - Applications



AI must fit into workflows.

ANATOMY / LAYER 3

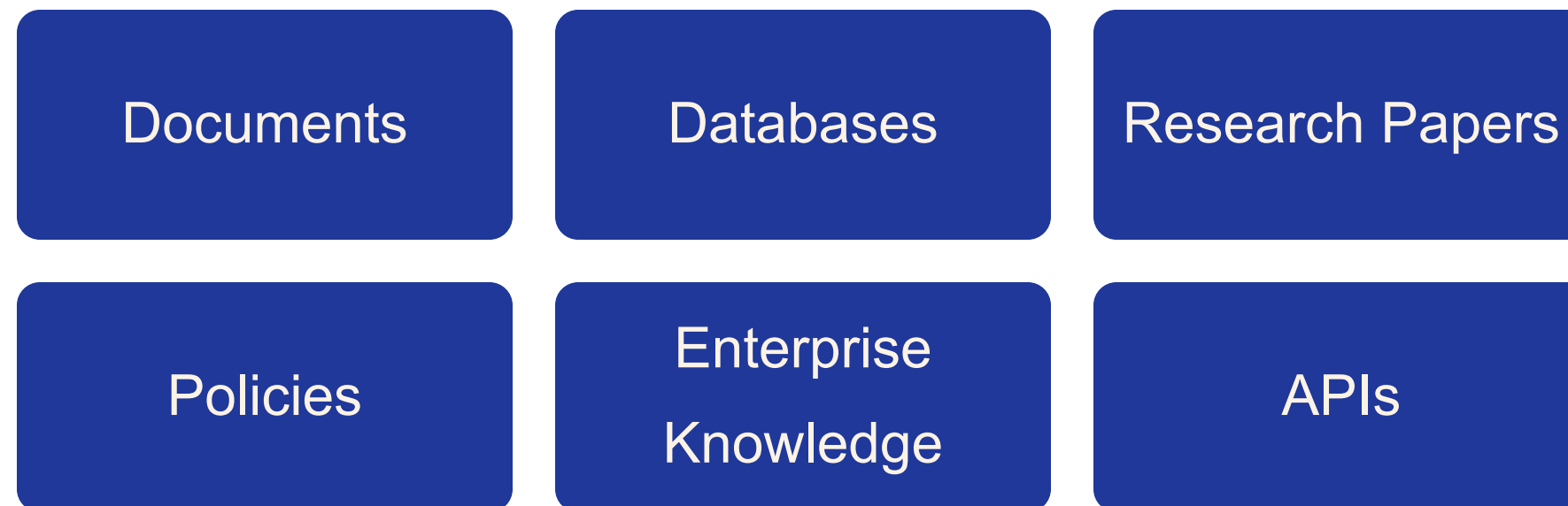
Layer 3 - AI Services



Modern AI systems combine multiple capabilities.

ANATOMY / LAYER 4

Layer 4 - Knowledge Layer



AI is limited by the quality of knowledge it can access.

ANATOMY / LAYER 5

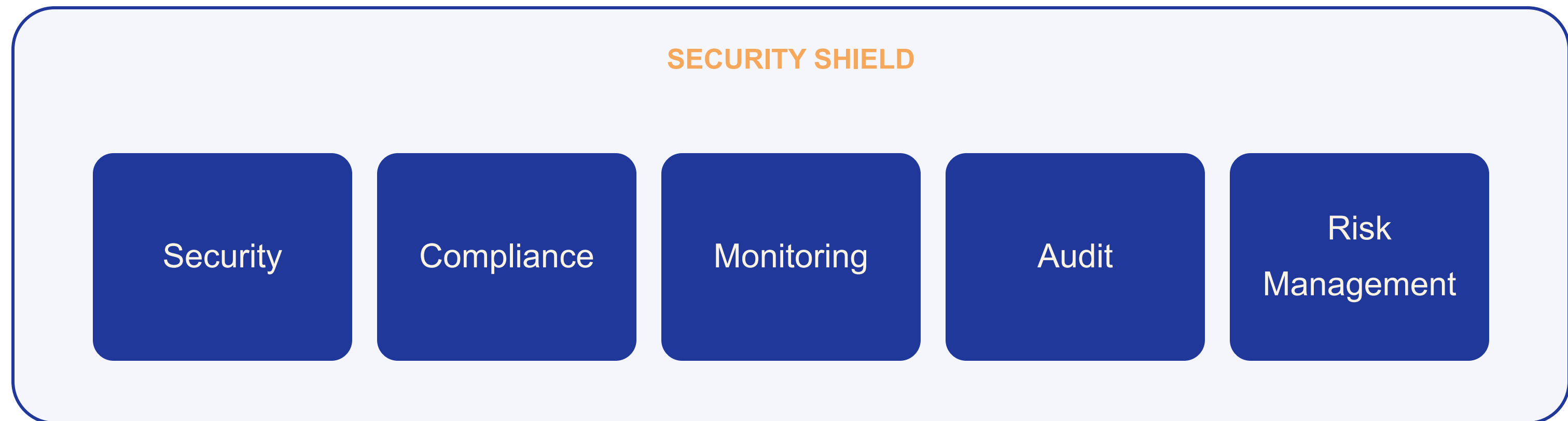
Layer 5 - Enterprise Systems



AI creates value when connected to operational systems.

ANATOMY / LAYER 6

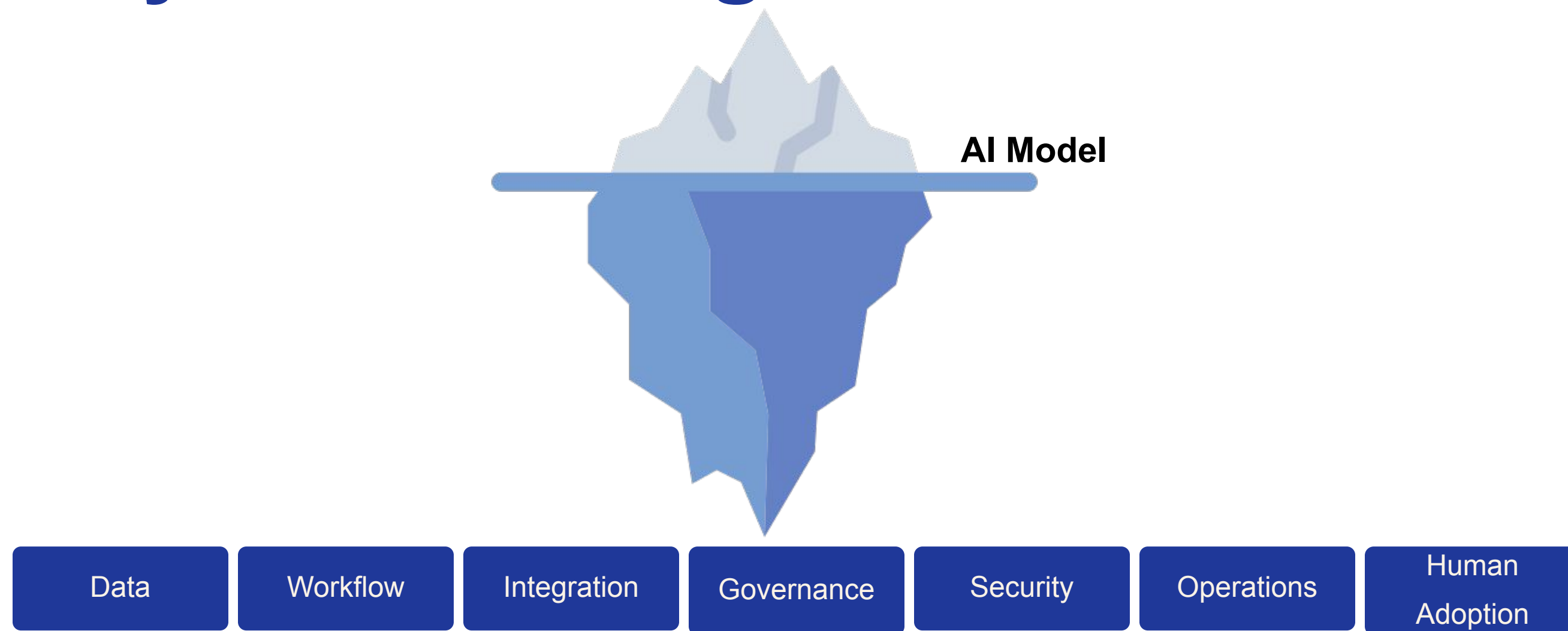
Layer 6 - Governance



Without governance, there is no production AI.

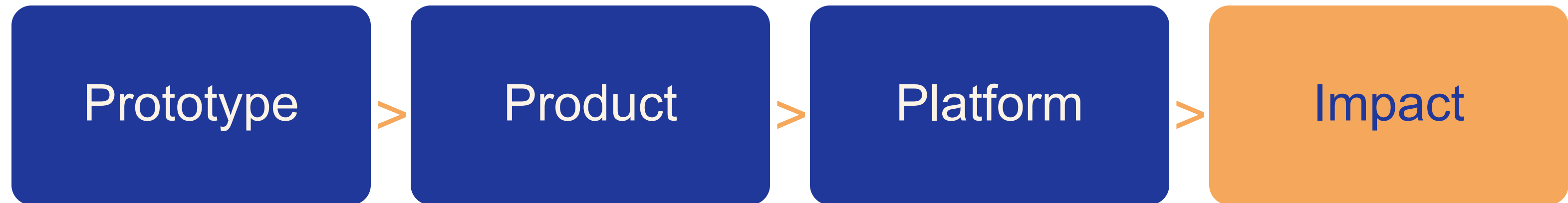
ANATOMY / COMPLEXITY

The AI System Iceberg



ANATOMY / THE JOURNEY

From Prototype to Impact



The goal is not to build a demo.

The goal is to create lasting impact.

TRANSITION / WHAT TO BUILD

Now That We Understand AI Systems...

What Problems Should We Solve?

Healthcare

Education

Digital Trade

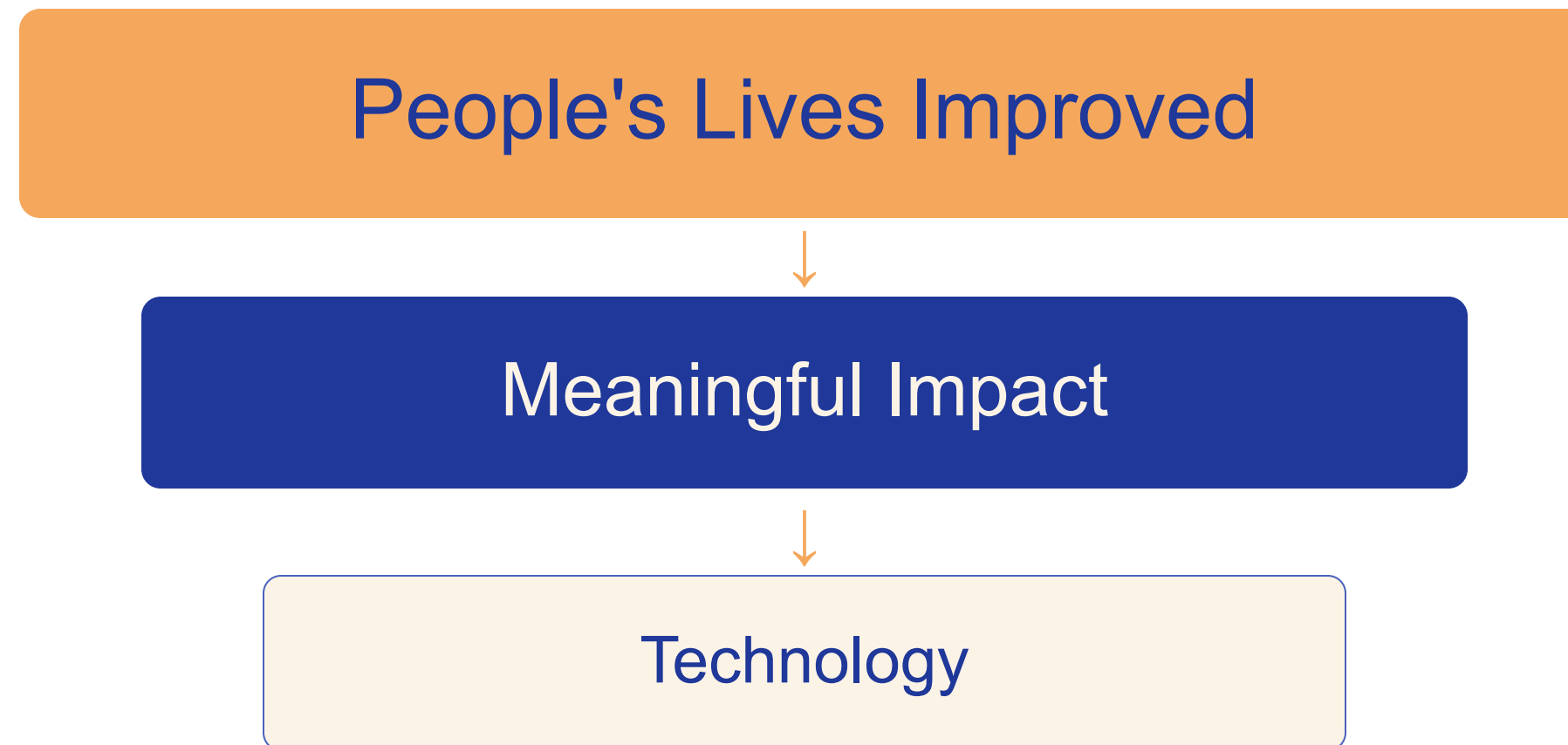
Climate

Agriculture

Next – Designing AI for Global Challenges

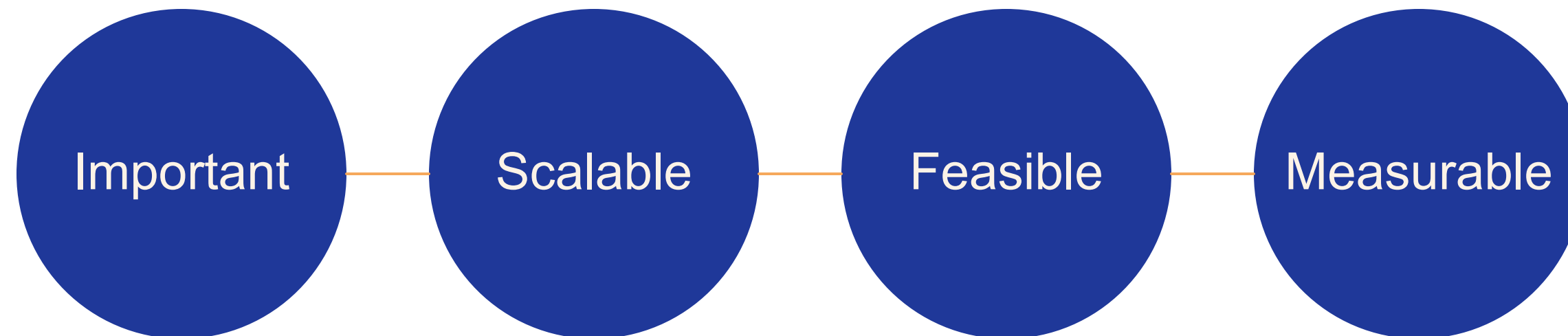
IMPACT / THE GOAL

Technology Is Not the Goal.



IMPACT / PROBLEM SELECTION

What Makes a Great Problem?



IMPACT / DIGITAL TRADE

Digital Trade.

**Millions of SMEs
Want to Sell Globally —
But Don't Know Where to Start.**

Export Readiness

Trade Advisory

Cross-Border
Commerce

Market Intelligence

Customer Support

IMPACT / PUBLIC SERVICES

Public Services.

**Government Services
Should Be Easier to Access.**

Citizen Assistant

Knowledge Assistant

Immigration Guidance

Public Information

Regulatory Navigation

IMPACT / HEALTHCARE

Healthcare.

**Healthcare Knowledge
Should Not Depend on Geography.**

Medical Information

Clinical Support

Rural Healthcare

Multilingual Guidance

IMPACT / EDUCATION

Education.

Knowledge Should Not Depend on Language.

AI Tutoring

Multilingual Learning

Accessibility

Personalized
Education

IMPACT / CLIMATE & RESILIENCE

Climate & Resilience.

**Better Decisions
Start With Better Information.**

Disaster Response

Climate Risk

Resource Optimization

Environmental
Monitoring

IMPACT / THE FLOW

From Problem to Impact.



Impact comes from solving problems —
not from deploying models.

TRANSITION / THE CALL

The World Does Not Need
More AI Demos.

The World Needs
More Solutions.

Next - Responsible AI in Practice

RESPONSIBLE AI / FOUNDATION

Responsible AI Is Not About Compliance

Responsible AI

=

Building Systems
People Can Trust

RESPONSIBLE AI / THE QUESTION

Just Because We Can Build It...

Doesn't Mean We Should

Build

or

Don't Build

RESPONSIBLE AI / QUESTION 1

Should We Build It?

Question #1

Facial Recognition

Mass Surveillance

Deepfakes

Autonomous Decisions

RESPONSIBLE AI / QUESTION 2

Who Benefits?

Question #2

Citizens

Customers

Governments

Businesses

Communities

RESPONSIBLE AI / QUESTION 3

Who Could Be Harmed?

Question #3

Bias

Exclusion

Misinformation

Privacy

Safety

RESPONSIBLE AI / REALITY

AI Failure Case Study

Building AI is easy.

Predicting human behavior is hard.

Biased
Hiring Systems

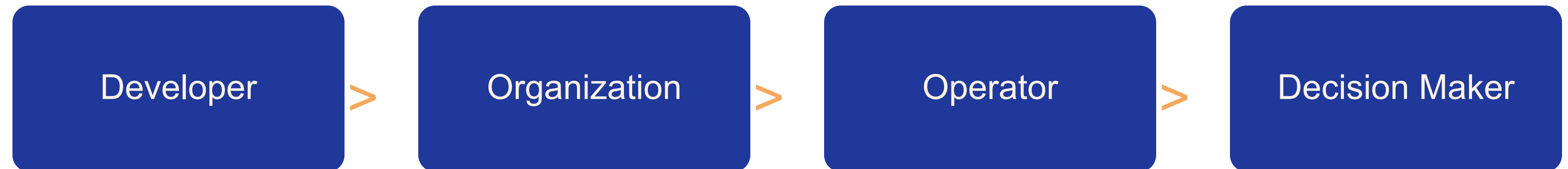
Misinformation
Bots

Unsafe
Recommendations

RESPONSIBLE AI / QUESTION 4

Who Is Accountable?

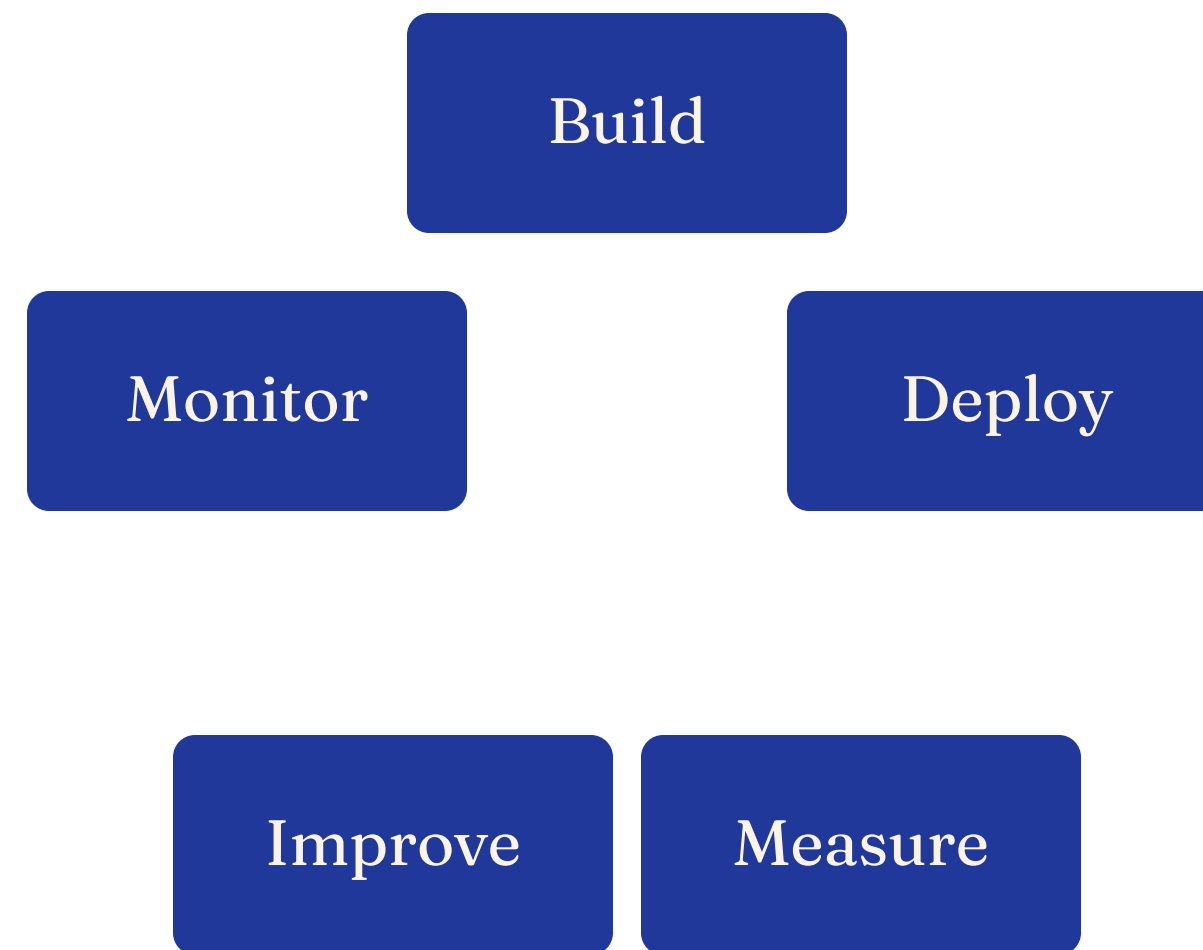
Question #4



RESPONSIBLE AI / QUESTION 5

How Do We Monitor It?

Question #5



RESPONSIBLE AI / CHECKLIST

Responsible AI Checklist

- ☒ Useful
- ☒ Fair
- ☒ Explainable
- ☒ Safe
- ☒ Accountable

TRANSITION / TRUST

Trust Is a Feature.

Without trust, there is no adoption.

Without adoption, there is no impact.

Next - How to Win This Hackathon

BUILD / PROBLEM FIRST

The Best AI Projects Start With a Problem.

Don't Start With AI



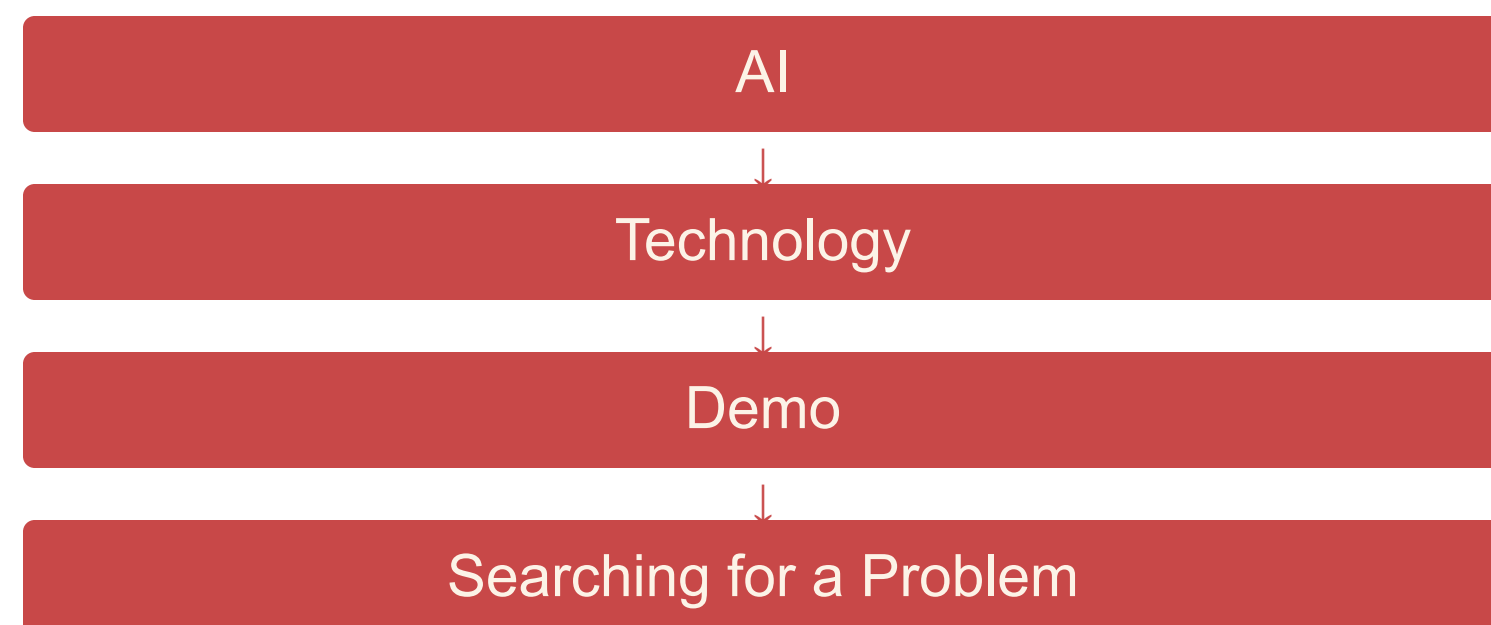
Start With a Problem

Great solutions begin with real human needs.

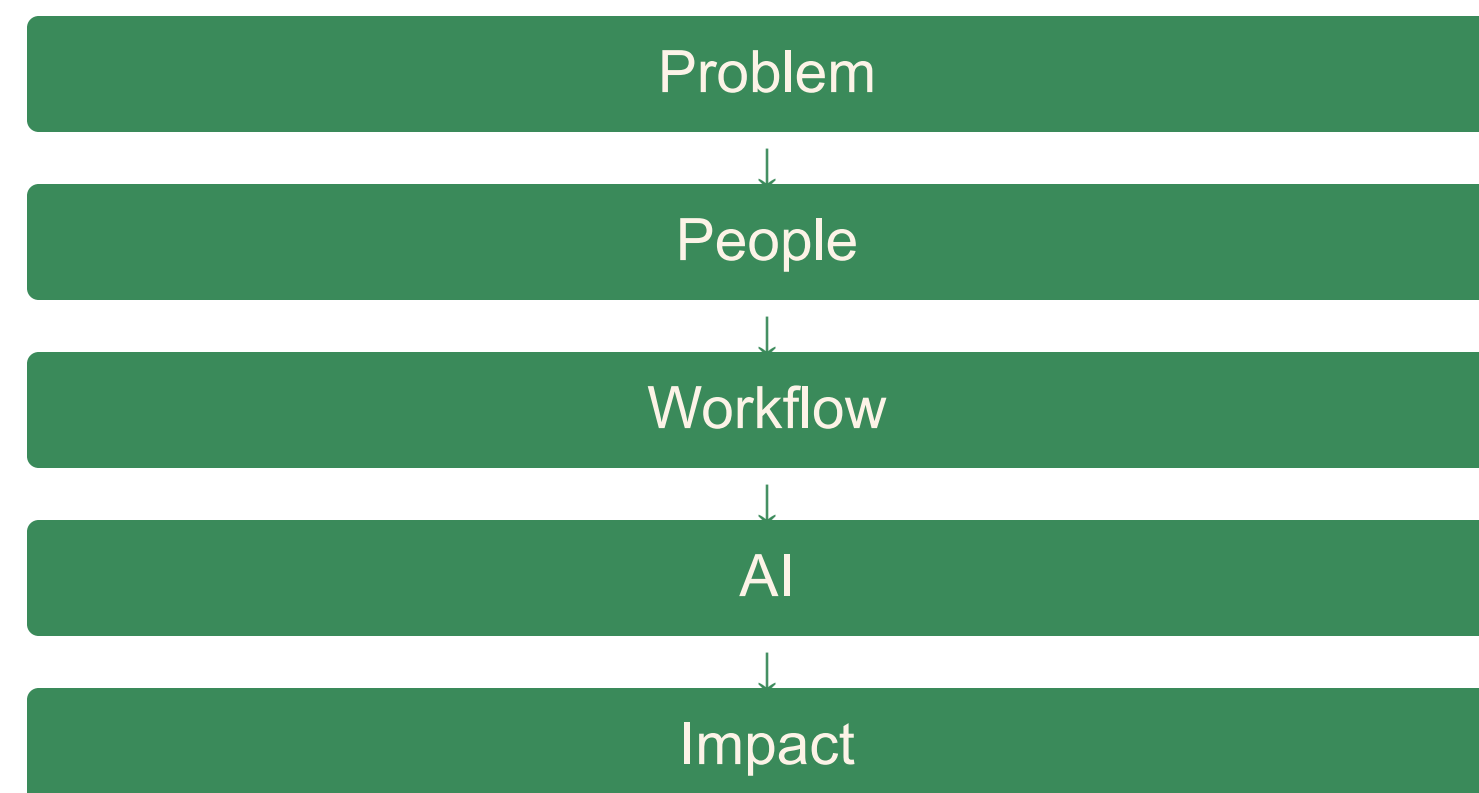
BUILD / WRONG VS RIGHT

The Trap Most Teams Fall Into.

Wrong



Right



BUILD / EVALUATION

What Judges Usually Look For.

Impact

Feasibility

Scalability

Innovation

Adoption

The best solution is rarely the most complicated.

BUILD / IMPACT

Impact.

Who Will Be Better Off Because This Exists?

Impact is about improving lives.

BUILD / FEASIBILITY

Feasibility.

Can This Actually Be Built?



A realistic solution beats a perfect fantasy.

BUILD / SCALABILITY

Scalability.

Can It Help

10
people

1,000
people

1 Million
people

Think beyond the prototype.

BUILD / INNOVATION

Innovation.

Innovation Is Not New Technology.

Innovation Is New Value.

Sometimes simple ideas create the biggest impact.

CLOSING / BUILD WHAT MATTERS

Build Something That Matters.

**The Future Belongs
To Those Who Can Turn AI
Into Real-World Impact.**

Thank you.